



E C E 222 Lec-002 Fall 2026 Syllabus

GENERAL INFORMATION

Course Subject, Number and Title

ECE 222: Electrodynamics I

Credits

4 credits

Course Description

Vector calculus application to electrodynamics problems; potential theory; static and dynamic electric and magnetic fields; macroscopic theory of dielectric and magnetic materials; Maxwell's equations; boundary conditions; wave equation; introduction to transmission lines.

Course Designations and Attributes

Undergraduate level course

Prerequisites

(PHYSICS 202, 208, or 248) and (MATH 234 or 376) or member of Engineering Guest Students. Not open to students with credit in E C E 220.

Credit Hours Fulfillment

This class meets for three 75-minute class periods each week over the fall semester and carries the expectation that students will work on course learning activities (reading, problem sets, studying, etc.) for about 3 hours for every class period. This syllabus includes more information about meeting times and expectations for student work.

LEARNING OUTCOMES

1. Apply concepts from single-variable and vector calculus to solve electrodynamics problems.
2. Calculate static electric and magnetic fields from charge and current distributions.
3. Use concepts of electric and magnetic fields to calculate forces, work, and energy.
4. Use boundary conditions on electric and magnetic fields to compute field changes at interfaces between materials.

5. Develop expressions for macroscopic electric circuit lumped parameters such as inductance and capacitance by applying principles of electric and magnetic fields.
6. Apply bounce diagrams to calculate time and space distribution of currents and voltages on transmission lines.

MEETING AND INSTRUCTOR INFORMATION

Meeting Time and Location

Monday, 4:00PM - 5:15PM	Engineering Hall 1227	In-person lecture
Wednesday, 4:00PM - 5:15PM	Engineering Hall 1227	In-person lecture
Friday, 4:00PM - 5:15PM	Engineering Hall 1227	In-person lecture

Instructional Modality

In-person, except when the instructor is not in Madison due to unexpected issues.

Instructor Contact Info

- **Name:** Prof. Haihan Sun
- **Office hours:**

Monday, 5:15PM - 6:00PM	Engineering Hall 3441
Wednesday, 5:15PM - 6:00PM	Engineering Hall 3441
Friday, 5:15 PM - 6:00 PM	Engineering Hall 3441

- **Email:** haihan.sun@wisc.edu

If you need additional help outside of office hours, please email Prof. Sun and include "[ECE 222]" in the subject line of your email.

Teaching Assistants (TAs) Contact Info

- **TA Name:** Kaimeng Liu & Shwetambari Sanjay Mali
- **Office hours:**

Tuesday, 1:00 to 2:00 PM	EH 2421	Kaimeng
Thursday, 1:00 to 2:00 PM	EH 2421	Kaimeng
Friday, 12:00 to 1:00 PM	EH 2421	Kaimeng
Wednesday, 12:00 - 1:00 PM	EH1422	Shwetambari

- **Email:** kliu377@wisc.edu and ssmali@wisc.edu

Please include "[ECE 222]" in the subject line of your email.

- **Office hours via Piazza:** Your instructor and TAs will also be available on **Piazza**, available through the Canvas site to answer text-based questions. Students are encouraged to post both questions and answers.

Overview

Course Website and Digital Instructional Tools

The course website is canvas.wisc.edu. All the important course announcements, homework assignments, homework solutions, supplementary materials (if any), and exams will be posted on this website. Primary methods of communications and making announcements in class is via Canvas (the Announcements Section) and via email. The instructor and the TA will use the email address that you have on file with UW to communicate with you. Please make sure to monitor your email account for important announcements and updates about ECE 222.

Required Textbook, Software and Other Course Materials

Textbook: F. Ulaby & U. Ravaioli, Fundamentals of Applied Electromagnetics, 8th edition, Pearson, 2020 (ISBN: 9780135200445).

This textbook is highly recommended. Any recent edition is okay, as homework will be posted separately on Canvas. There are digital and print versions available from the publisher at the following website, and you can find this textbook or other recent editions new or used from a variety of sources.

Textbook website: <https://www.pearson.com/en-us/subject-catalog/p/fundamentals-of-applied-electromagnetics/P200000003174/9780135200445>

Homework and Other Assignments

Homework:

- Approximately 11 homework sets will be assigned during this semester.
- Homework must be submitted electronically via Canvas. Each homework assignment should be submitted in the form of a single PDF document. The file should be named as "HW_No_X_Last Name.pdf". Please replace 'X' with the number of the homework assignment and replace "Last Name" with your own last name.
- Homework submissions should be neatly written and well-organized, with all steps clearly shown and final answers boxed for easy identification.
- Homework sets are graded based on completion. The lowest grade will be dropped in the final grade calculation.
- **Late homework assignments will not be accepted. Individual homework extensions will not be granted under normal circumstances.**

In-class Example Problems:

- On Wednesdays/Mondays, we will do in-class example problems in small groups and all together.
- These must be electronically uploaded to Canvas by **11:59 pm on the following day (Thursdays/Tuesdays)**.

- They are graded on a completion basis. The lowest grade will be dropped in the final grade calculation.
- Late In-Class Problem submission will not be accepted. Individual extensions will not be granted under normal circumstances.

Exams

- There will be two in-class exams and one final exam.

General Guidelines for Exam Proctoring

All exams will be proctored by the instructor and/or the TA or the instructor's designated proctor.

Grading

In-class Problems	100 pts
Homework	100 pts
Midterm Exam 1	100 pts
Midterm Exam 2	100 pts
Final Exam	100 pts
Total Points	500 pts

Tentative Grading Scale (No curve)

>90% A
 86-89 AB
 80-85 B
 76-79 BC
 68-75 C
 60-67 D
 <60 F

Tentative Schedule

Schedule may be subject to change.

Week	Class #	Date	Day	Reading	Topics	Due
1	1	Sept. 2	W	Ch. 3	Review of vector calculus	
	2	Sept. 4	F			
2		Sept. 7	M		Labor day (no class)	
	3	Sept. 9	W	4.1 – 4.4	Electrostatics	HW 1
	4	Sept. 11	F			
3	5	Sept. 14	M	4.5	Electric potential	
	6	Sept. 16	W			HW 2
	7	Sept. 18	F			
4	8	Sept. 21	M	4.6, 4.7	Electrical properties of materials	

	9	Sept. 23	W	6.9, 6.10	(conductors & dielectrics)	HW 3
	10	Sept. 25	F			
5	11	Sept. 28	M			
	12	Sept. 30	W		Review	HW 4
	13	Oct. 2	F		Midterm exam 1, in class	
6	14	Oct. 5	M	4.8 – 4.10	Electrostatic boundary conditions, Energy stored in electrostatic field, Capacitance	
	15	Oct. 7	W			
	16	Oct. 9	F			
7	17	Oct. 12	M	5.1, 5.2	Magnetostatic Maxwell's equations, Force and torque in magnetic field, Biot-Savart's law	
	18	Oct. 14	W			HW 5
	19	Oct. 16	F			
8	20	Oct. 19	M	5.3, 5.6	Ampere's law, Magnetic boundary conditions	
	21	Oct. 21	W			HW 6
	22	Oct. 23	F			
9	23	Oct. 26	M	5.5, 5.7, 5.8	Magnetic properties of materials, Self-inductance, Magnetic energy	
	24	Oct. 28	W			HW 7
	25	Oct. 30	F			
10	26	Nov. 2	M			
	27	Nov. 4	W		Review	HW 8
	28	Nov. 6	F		Midterm exam 2, in class	
11	29	Nov. 9	M	6.1 – 6.7	Electromagnetic induction	
		Nov. 11	W		No class	
		Nov. 13	F		No class	
12	30	Nov. 16	M	6.1 – 6.7; 7.1 – 7.3	Electromagnetic induction Displacement current density, Transformer Intro to EM waves	
	31	Nov. 18	W			HW 9
	32	Nov. 20	F			
13	33	Nov. 23	M			
	34	Nov. 25	W			
	35	Nov. 27	F		No class (Thanksgiving recess)	
14	36	Nov. 30	M	7.1 – 7.3, 8.1.1– 8.1.3	Maxwell's equations in phasor form Transmission lines & transients	
		Dec. 2	W			HW 10
	37	Dec. 4	F			
15	38	Dec. 7	M	Ch. 2	Bounce diagrams Review Session	
	39	Dec. 9	W			HW 11
16		Dec. 11	F		Final Exam (5:05-7:05 pm), Location: TBD	

Academic Policies

Academic Integrity

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin–Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of disciplinary sanctions include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension or expulsion.

Use of Artificial Intelligence (AI) Tools

You are welcome to use AI tools and applications (such as Copilot) to aid in your studying or understanding of homework problems and in-class examples. Remember that AI (and other digital tools and resources) will not be available during the exams, and over-reliance on AI will hinder your preparation. Any work that you turn in should be your own. For example, it is acceptable to use AI to understand how to approach a problem you are struggling with or how to solve a piece of a problem when you are stuck. However, you should not type a problem statement into an AI tool and copy the solution. Please be aware you are responsible for the information you submit based on an AI query. Your use of AI tools should be documented and cited. More information about use and documentation of AI can be found here:

<https://researchguides.library.wisc.edu/GenerativeAI/Home>

Accommodations for Students with Disabilities

The University of Wisconsin–Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12) and UW–Madison policy (UW-855) require the university to provide reasonable accommodations to students with disabilities to access and participate in its academic programs and educational services. Faculty and students share responsibility in the accommodation process. Students are expected to inform faculty of their need for instructional accommodations during the beginning of the semester, or as soon as possible after being approved for accommodations. Faculty will work either directly with the student or in coordination with the McBurney Disability Resource Center to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.

Engineering for All

We cultivate and uphold a safe, welcoming, and supportive environment for all, operating with the highest standards of integrity, collegiality, and ethics and recognizing the contributions of every member of our college community. Our faculty integrate innovative, multidisciplinary research into the classroom, leveraging

evidence-based teaching methods to inspire and prepare our students. Through our research, teaching and service, we pursue answers to societal needs and challenges.

Academic Calendar and Religious Observances

View the full [academic calendar](#) in addition to information about religious and election day observances. Students are responsible for notifying instructors within the first two weeks of classes about any need for flexibility due to [religious observances](#).

Establishment of the academic calendar for the University of Wisconsin–Madison falls within the authority of the faculty as set forth in [Faculty Policies and Procedures](#). Construction of the academic calendar is subject to various rules and laws prescribed by the Board of Regents, the Faculty Senate, State of Wisconsin and the federal government. Find [additional dates and deadlines for students](#) on the Office of the Registrar website.

Mental Health and Well-Being

As a student you may experience a range of issues that can cause barriers to learning. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, or loss of motivation. University Health Services can help with these or other issues you may experience. Help is always available. You can learn about free, confidential mental health services available to you; call 608-265-6600 (option 2) or visit uhs.wisc.edu. College of Engineering students can also directly access short-term mental health support from the engineering embedded mental health provider. Visit the [Engineering Wellness](#) page for more information.

Privacy of Student Records & Use of Audio Recorded Lectures

Lecture materials and recordings for this course are protected intellectual property at UW–Madison. Students enrolled in this course may use the materials and recordings for their personal use related to participation in the course. Students may also take notes solely for their personal use. If a lecture is not already recorded, students are not authorized to record lectures without permission unless they are considered by the university to be a qualified student with a disability who has an approved accommodation that includes recording. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities, with the exception of sharing copies of personal notes as a notetaker through the McBurney Disability Resource Center. Students are otherwise prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct. View [more information about FERPA](#).

Teaching & Learning Data Transparency

The privacy and security of faculty, staff and students' personal information is a top priority for UW–Madison. The university carefully reviews and vets all campus-supported digital tools used for teaching and learning, including those that support [data empowered educational practices](#) and proctoring. View more information about [teaching and learning data transparency](#) at UW–Madison.

Course Evaluations

Students at the University of Wisconsin–Madison have the opportunity to evaluate their learning experiences and the courses they are enrolled in through course evaluations. Many instructors use a digital course evaluation tool to collect feedback from students. Student participation is an integral component of course development, and confidential feedback is important. UW–Madison strongly encourages student participation in course evaluations.

Students' Rules, Rights & Responsibilities

View more information about [student rules, rights and responsibilities](#) such as student privacy rights, sharing of academic record information, academic integrity and grievances. Additional College of Engineering information can be found at: <https://engineering.wisc.edu/report-an-incident/academic-grievances-and-complaints/>